



MANIPAL ACADEMY OF HIGHER EDUCATION

DEPARTMENT OF MATHEMATICS
IV SEM B.TECH. MATHEMATICS AND COMPUTING
END SEMESTER EXAMINATION

Date of Examination: 08 May 2025

Time: 2:30pm-5:30pm

MAX.MARKS: 50

VECTOR ANALYSIS AND COMPLEX VARIABLES [MAT 2238]

Marks: 50

Duration: 180 mins.

A

Answer all the questions.

Section Duration: 180 mins

- 1) Answer all questions.
- 2) Missing data if any can be assumed with proper reasoning.
- 3) (Capital) $\text{Log}(z)$ and (capital) $\text{Arg}(z)$ will always denote the principal branch of the logarithm and the argument function

1a) Find the equation for the tangent plane to the surface $xyz = a^3$ at a general point (x_0, y_0, z_0) on the surface. Here $a \neq 0$. (3)

1b) Evaluate the directional derivative of $f(x, y, z) = 3x - 5y + 2z$ at $(2, 2, 1)$ in the direction of the outward normal to the sphere $x^2 + y^2 + z^2 = 9$. (4)

1c) Given the curve $C: r(t) = (10 \cos t, 1, 10 \sin t)$, find the parametric representation of the tangent line to C at $P = (6, 1, 8)$. (3)

2a) Let $\vec{v}: \mathbb{R}^3 \rightarrow \mathbb{R}^3$ be a twice continuously differentiable vector valued function. Evaluate $\text{div}(\text{curl } \vec{v})$. (3)

2b) Show that $u(x, y) = 2x(1 - y)$ is harmonic on $\mathbb{R}^2$ and find a harmonic conjugate of $u(x, y)$. (3)

2c) For $0 < \theta < 2\pi$, derive the identity

$$1 + \cos \theta + \cos 2\theta + \cdots + \cos n\theta = \frac{1}{2} + \frac{\sin[(2n+1)\theta/2]}{2 \sin \theta/2}$$

Here, $n \in \mathbb{N} \cup \{0\}$. (4)

Hint: Use the identity

$$1 + z + z^2 + \cdots + z^n = \frac{1 - z^{n+1}}{1 - z}, z \neq 1$$

3a) Find the values of a) $\text{Log}(-ei)$ and b) $\text{Log}(1-i)$. (3)

3b) Find the value of the limit L below and using the $\epsilon - \delta$ definition of limit, prove that

$$\lim_{z \rightarrow i} \frac{iz^3 - 1}{z + i} = L \quad (4)$$

3c) Evaluate the integral

$$\oint_C \text{Log}(1-z) dz \quad (3)$$

where C is the boundary of the parallelogram with vertices $\pm i, \pm(1+i)$. Provide accurate and sufficient justifications.

4a) Let C denote the line segment from $z = i$ to $z = 1$. Show that of all the points on that line segment, the midpoint is the closest to the origin and hence, prove that

$$\left| \int_C \frac{dz}{z^4} \right| \leq 4\sqrt{2} \quad (3)$$

4b) Let f be analytic on a simply connected domain D and let C be a simple closed contour (5)
 C inside D . If z_0 is not on C (note that z_0 could be in the interior of C as well as exterior), then, show that

$$\oint_C \frac{f'(z)}{z - z_0} dz = \oint_C \frac{f(z)}{(z - z_0)^2} dz$$

4c) Evaluate $\int_C (\bar{z})^2 dz$ where C is the straight line from 0 to 2+i. (2)

5a) Let $f: \mathbb{R}^3 \rightarrow \mathbb{R}^2$ be a vector field defined as

$$f(x, y, z) = (x^2 + y + z)i + (2x + y + z^2)j. \quad (3)$$

Compute the Jacobian matrix of f with respect to (x, y, z) .

5b) Let

$$K = f(R) = R^T R$$

where R is an 3 x 3 matrix with real entries. Find the partial derivative of

- $(2,3)^{\text{th}}$ component of K, say $K_{2,3}$ with respect to $R_{1,2}$
- $(3,1)^{\text{th}}$ component of K, say $K_{3,1}$ with respect to $R_{2,1}$
- $(3,3)^{\text{th}}$ component of K, say $K_{3,3}$ with respect to $R_{1,3}$.

(3)

5c) Consider a simple neural network with an input layer of 2 neurons ($x_1 = 0.5, x_2 = 0.3$) and output layer with 1 neuron (target output $y = 1.0$) activated by the sigmoid function $\sigma(x) = 1/(1 + e^{-x})$. Assuming the loss function to be the mean square error function $C(a^1) = \frac{1}{2}(a^1 - y)^2$ and the learning rate to be 0.4

(4)

and initial parameters to be $w_1 = 0.4, w_2 = 0.6, b = 0.2$, perform one step each of forward propagation and backward propagation using gradient descent and present all the computed values (including intermediate variable z , predicted value $a^{(1)}$, all gradients and updated parameter values) in a table. Round off your values to 4 decimal places.

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